Fundamentos de Algoritmos e Estrutura de Dados – Aula 02
Complexidade e Programação Dinâmica

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Plano de Aula

- **Complexidade Computacional**
- **Notação BigO**
- **Programação Dinâmica**

Tópico 01 – Complexidade e Notação Assintótica (BigO)

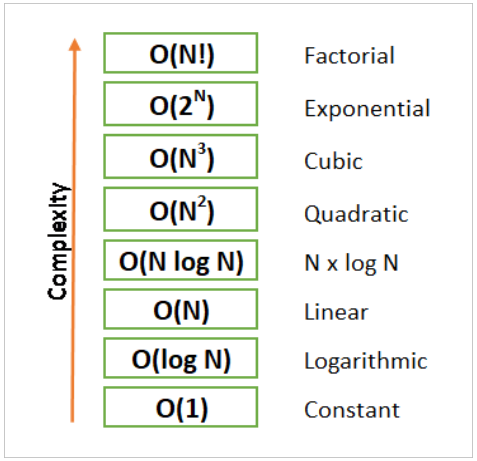
Complexidade Computacional

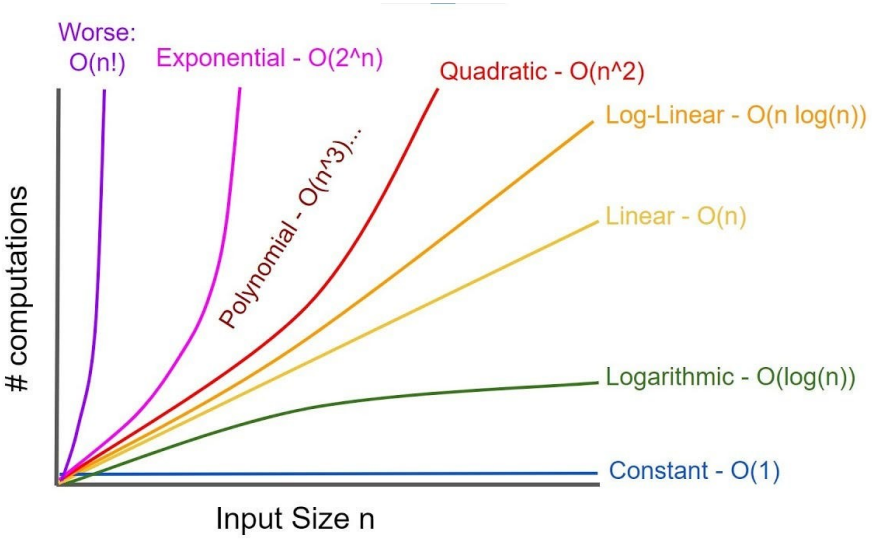
- **Complexidade Computacional mede o esforço de um algoritmo em computar um dado**

- **Problema**
  - **Qual o custo para encontrarmos um valor em um vetor de tamanho N?**

Notação BigO

- **Análise Assintótica (Problemas Grandes)**
- **BigO → O( )**
  - **Formalização da complexidade de um algoritmo em razão da sua entrada (N).**
  - **Tempo ou Recurso em função de N**
  - **Considera-se o pior cenário para N (tende ao infinito)**

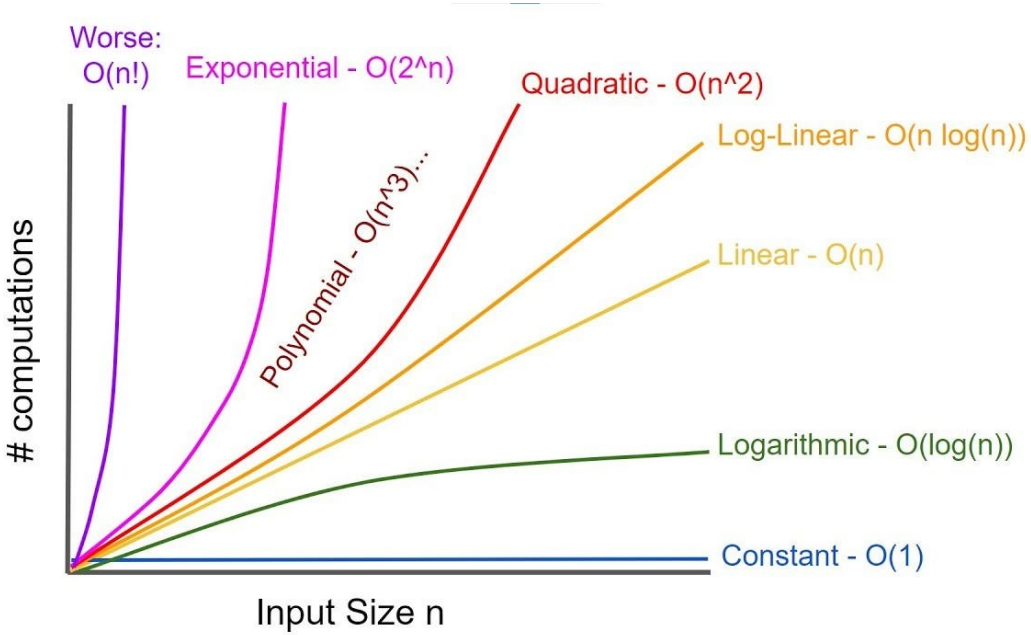




Constante: O(1)

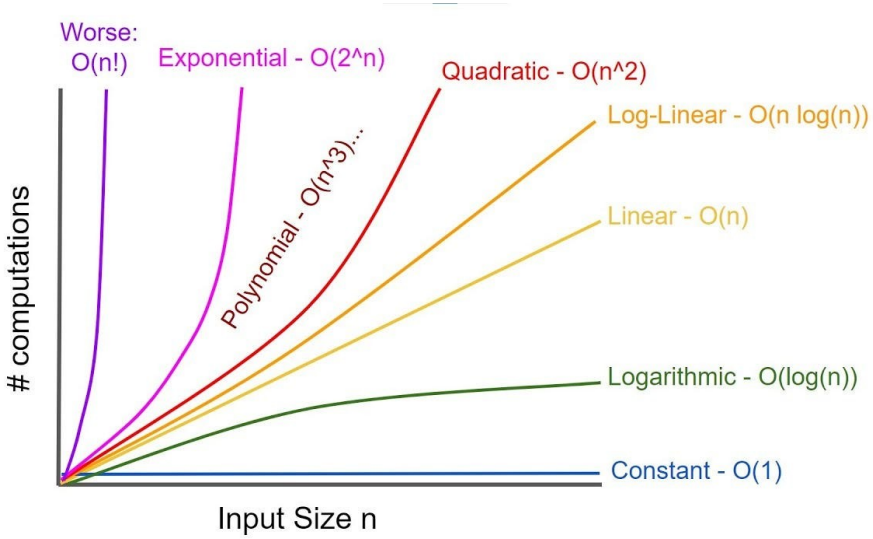
```c
void printFirstElementOfArray(int arr[])
{
    printf("First element of array = %d",arr[0]);
}
```

- **N = 1 → 1 iter → O(1)**
- **N = 1000 → 1 iter → O(1)**
- **N= 1000000 → 1 iter → O(1)**



Linear: O(n)

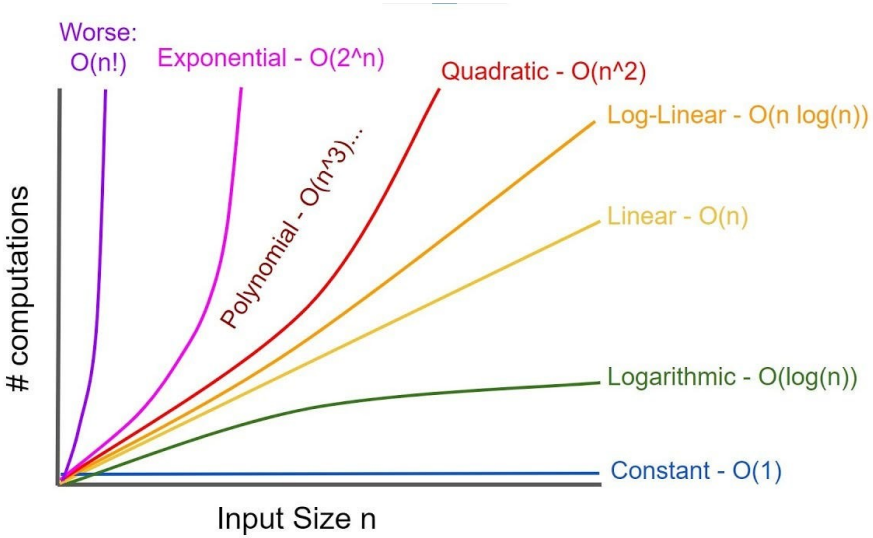
```c
void printAllElementOfArray(int arr[], int size)
{
    for (int i = 0; i < size; i++)
    {
        printf("%d\n", arr[i]);
    }
}
```



- **N = 1 → 1 iter → O(N)**
- **N = 1000 → 1000 iter → O(N)**
- **N= 10000 → 10000 iter → O(N)**

Quadrática: $O(n^2)$

```c
void printAllPossibleOrderedPairs(int arr[], int size)
{
    for (int i = 0; i < size; i++)
    {
        for (int j = 0; j < size; j++)
        {
            printf("%d = %d\n", arr[i], arr[j]);
        }
    }
}
```

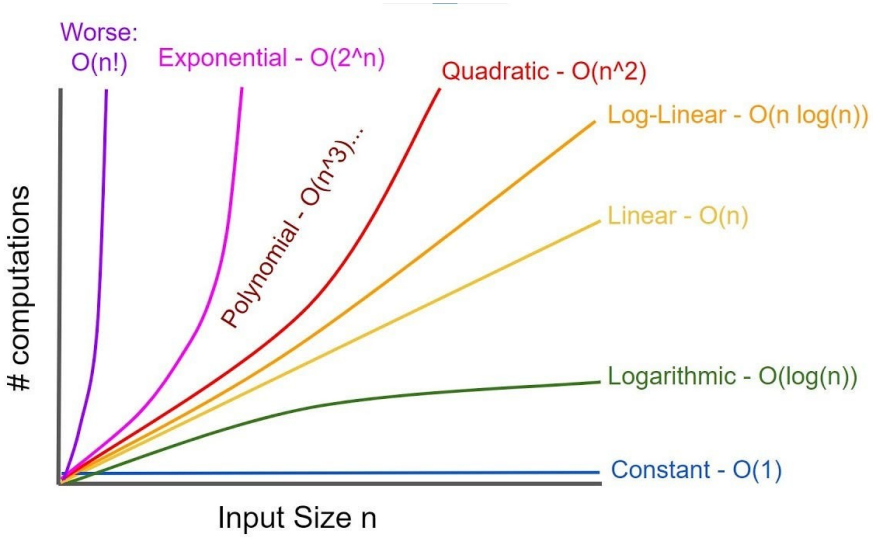


- **N = 2 → 4 → $0(n^2)$**
- **N = 4 → 16 → $0(n^2)$**
- **N= 100 → 10000 → $O(n^2)$**

- **Cubica → $O(n^3)$**
  - **3 laços aninhados**

Logaritmica: O(log n)

```python
items = [1, 2, 3, 4, 5, 6 , 7, 8, 9, 10]

def binary_search(alist, item):
  first = 0
  last = len(alist)-1
  found = False

  while first <= last and not found:

    midpoint = (first + last)//2

    if alist[midpoint] == item:
      found = True
    else:
      if item < alist[midpoint]:
        last = midpoint-1
      else:
        first = midpoint+1

    return found

print(binary_search(items, 19))
```



- **N = 2 → 1 → 0(log n)**
- **N = 4 → 2 → 0(log n)**
- **N= 100 → 10 → O(log n)**

O(n log n)

```
for(int i = 0; i < n; i++)
  for(int j = 1; j < n; j = j * 2)
      print(i,j)
```

- **0(n) * 0(log n) == 0(n log n)**

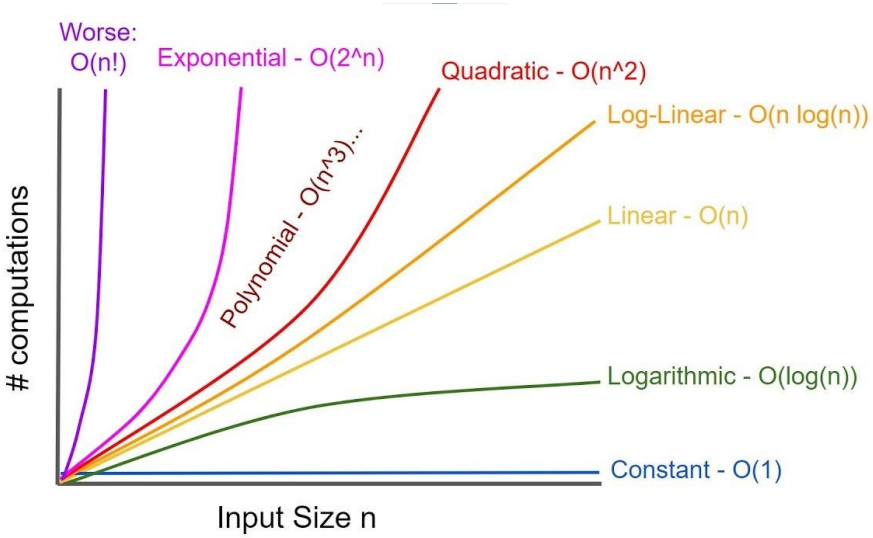
N=2 -> 2

N=4 -> 8

N=6 -> 18

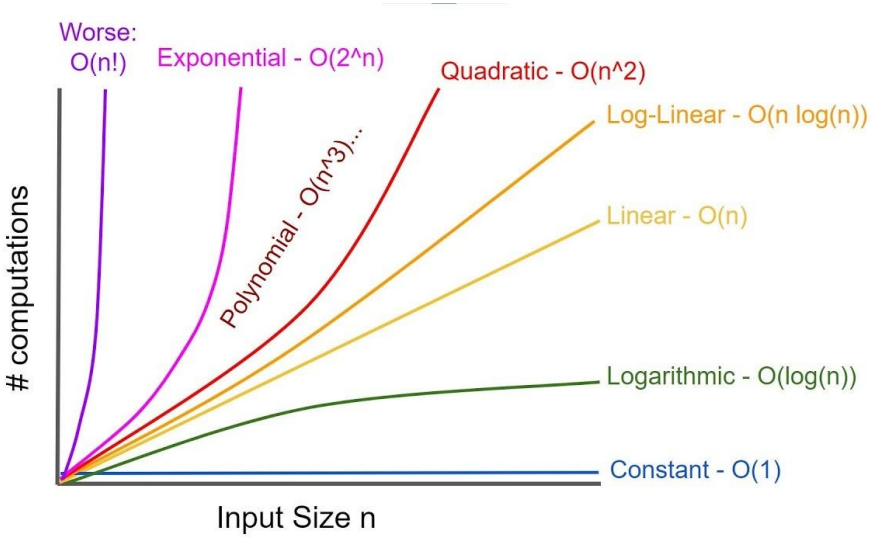
N=8 -> 24

N=10 -> 40



Exponencial: $O(2^n)$

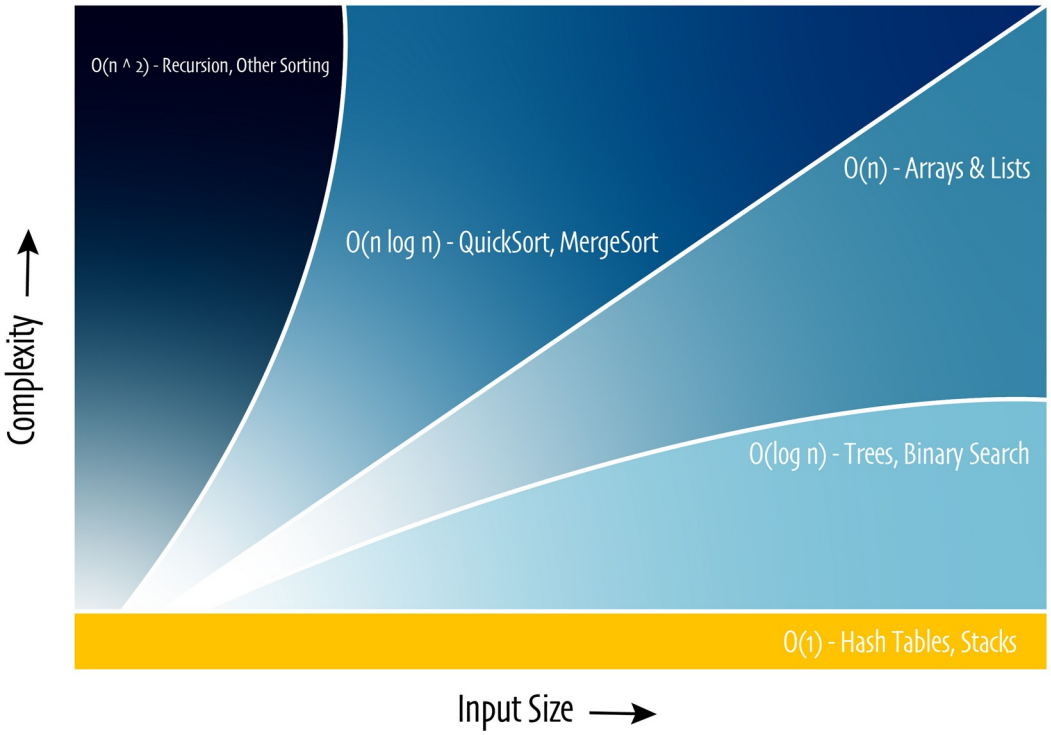
```
int fibonacci(int num)
{
    if (num <= 1) return num;
    return fibonacci(num - 2) + fibonacci(num - 1);
}
```



- **N = 2 → 4 → $0(2^n)$**
- **N = 4 → 256 → $0(2^n)$**
- **N= 10 → 1024 → $O(2^n)$**

Big O – Estruturas de Dados

BIG O - Buscas/Ordenação



Dicas ao computar BigO

- **Elimine Constantes**

O(2n) → O(n)

```
void printAllItemsTwice(int arr[], int size)
{
    for (int i = 0; i < size; i++)
    {
        printf("%d\n", arr[i]);
    }

    for (int i = 0; i < size; i++)
    {
        printf("%d\n", arr[i]);
    }
}
```

O(1 + n/2 + 100) → O(n)

```
void printFirstItemThenFirstHalfThenSayHi100Times(int arr[], int size)
{
    printf("First element of array = %d\n",arr[0]);

    for (int i = 0; i < size/2; i++)
    {
        printf("%d\n", arr[i]);
    }

    for (int i = 0; i < 100; i++)
    {
        printf("Hi\n");
    }
}
```

Dicas ao computar BigO

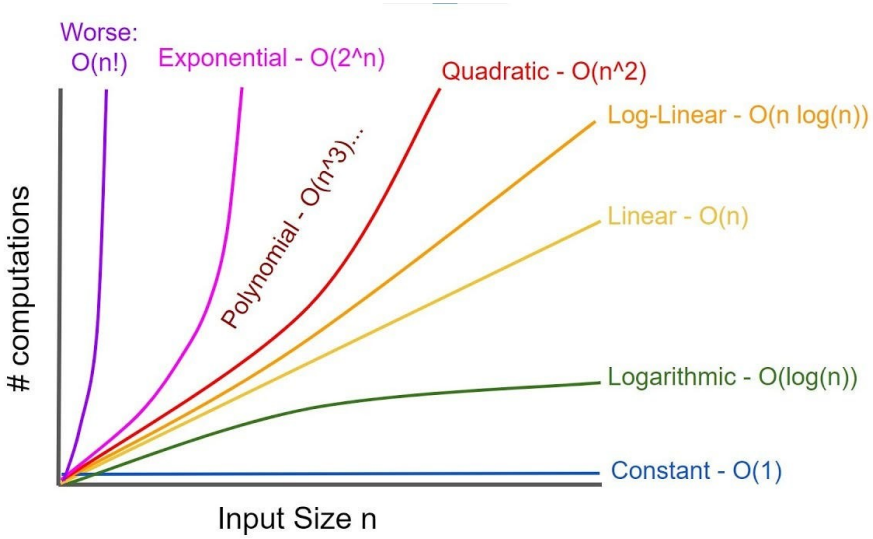
- **Ignore os termos/passos de menor relevância**
- **Ex:**
  - **$O(n + n^2) \rightarrow O(n^2)$**

```
for (int i = 0; i < size; i++)
{
    printf("%d\n", arr[i]);
}

for (int i = 0; i < size; i++)
{
    for (int j = 0; j < size; j++)
    {
        printf("%d\n", arr[i] + arr[j]);
    }
}
```

  - **$O(n^3 + 50n^2 + 10000) \rightarrow O(n^3)$**
  - **$O((n + 30) * (n + 5)) \rightarrow O(n^2)$**

Linear: O(n)

```c
void printAllElementOfArray(int arr[], int size)
{
    for (int i = 0; i < size; i++)
    {
        printf("%d\n", arr[i]);
    }
}
```



- **N = 1 → 1 iter → O(N)**
- **N = 1000 → 1000 iter → O(N)**
- **N= 10000 → 10000 iter → O(N)**

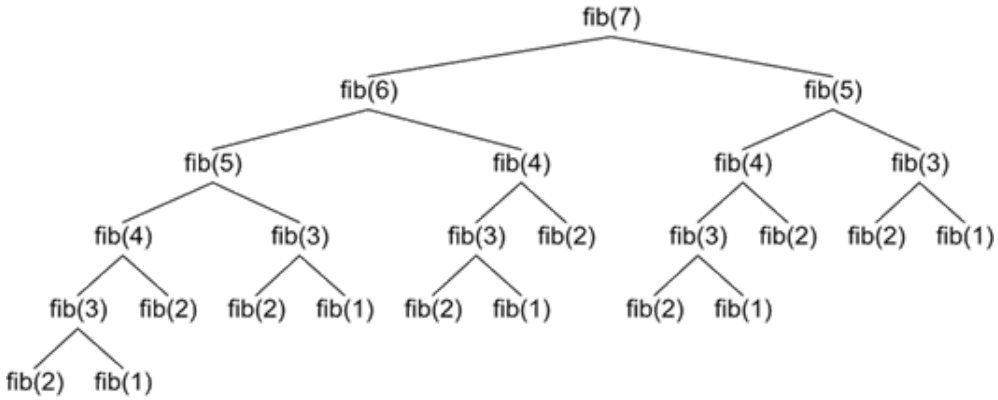
Tópico 02 – Programação Dinâmica

Programação Dinâmica

- **Richard Belmamm , 1950**
- **‘Programação’ tem sentido de ‘Planejamento’**
- **Problemas recursivos com sobreposição de subproblemas**
- **Utiliza uma estrutura computacional (vetor/matriz) para armazenar resultados dos sub-problemas**
- **Ex: Fibonacci recursivo é ineficiente (Exponencial $O(2^n)$))**

$$F(n) = \left\{ \begin{array}{cc} 1 & \text{se } n \leq 1 \\ F(n-1) + F(n-2) & \text{caso contrário} \end{array} \right\}$$

```
int fibonacci(int num)
{
    if (num <= 1) return num;
    return fibonacci(num - 2) + fibonacci(num - 1);
}
```



Programação Dinâmica

- **Solução: Armazenar subproblemas já computados**
- **O(n)**

```
def FibonacciDP(n,computed={0:0,1:1}):

  if n not in computed:
    computed[n] = FibonacciDP(n-1,computed) + FibonacciDP(n-2,computed)

  return computed[n]
```

- **Codificar!**